

Introduction to Data Science

Data Acquisition & Web Scraping Logic

Dept. CSE

Week 2: Class 2

Learning Objectives

By the end of this class, students will be able to:

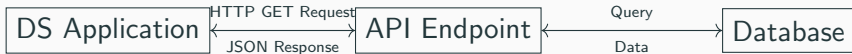
- Explain the logic of **APIs** (Application Programming Interfaces) as a data source.
- Contrast **Web Scraping** with **Web Crawling**.
- Evaluate the ethical and legal boundaries of data acquisition.

How do we "Obtain" Data?

In the OSEMN model, "O" is Obtain. In CSE, we don't usually get a nice CSV file. We have to fetch it.

1. **Direct Database Access:** SQL Queries (Internal).
2. **APIs:** Structured requests to a server (External).
3. **Web Scraping:** Extracting data from HTML (When no API exists).

The Logic of an API



- **The Contract:** An API defines exactly what you can ask for and how it will be delivered.
- **Authentication:** API Keys ensure only authorized data scientists get the data.

Web Scraping: The Last Resort

The Logic:

- Every website is just HTML (Semi-structured).
- We use logic to find patterns in tags: e.g., all prices are inside `<span>` tags with class "price".

Challenges:

- **Fragility:** If the website designer changes the color, the scraper might break.
- **Politeness:** Scraping too fast can crash a server (Denial of Service).

- **Robots.txt:** A file on every website telling bots what they *cannot* scrape.
- **Terms of Service (ToS):** Legally, if you "click accept," you might be agreeing not to scrape data.
- **Public vs. Private:** Just because data is on the web doesn't mean it's "Public Domain."

Assessment (Bloom's C3 - Application)

Task: API Request Logic

A Weather API requires the following URL structure:

`api.weather.com/get?city=[CITY_NAME]&key=[YOUR_KEY]`

Problem:

1. Write the hypothetical URL to fetch weather data for "Dhaka" using the key "ABC123".
2. If the API returns a "404 Error", based on your CSE knowledge, what does this imply about your data acquisition step?

Assessment (Bloom's C4 - Analysis)

Comparison: API vs. Scraping You are tasked with getting real-time stock prices for a Data Science project.

1. Analyze why an **API** is superior to **Scraping** for this task in terms of *Veracity* (Data Quality).
2. Suppose you scrape a news site for "Sentiment Analysis." The site blocks your IP address after 100 requests. Design a logic (without code) to prevent this from happening (e.g., timing, headers).
3. **Ethical Analysis:** Is it ethical to scrape a competitor's pricing data every hour to automatically lower your own prices? Justify using the "Value" vs "Veracity" principles.

OLTP vs. OLAP

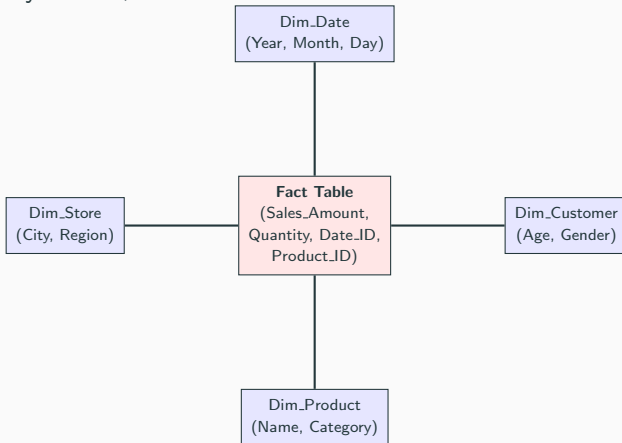
Feature	Database (OLTP)	Data Warehouse (OLAP)
Purpose	Day-to-day operations	Decision making / Analysis
Data State	Current, highly normalized	Historical, de-normalized
Query Type	Simple (Insert/Update)	Complex (Aggregations/Sums)
Optimization	Fast writes	Fast reads

The DS Connection

Data Scientists rarely query OLTP directly (to avoid crashing the app). We work in the **Warehouse**.

Multidimensional Modeling: The Star Schema

To make analysis fast, we use a **Star Schema** instead of 3rd



Normal Form.

- **Facts:** Quantitative data (STA 101 variables).
- **Dimensions:** Contextual attributes (for grouping/filtering).

Schema-on-Write Logic

- In a Warehouse, the structure (Schema) must be defined **before** data is loaded.
- **Math Logic:** If you add a "Price" column as an *Integer* but the data has *Decimals*, the system rejects it.
- This ensures **Veracity** but reduces **Flexibility**.

Assessment (Bloom's C3 - Application)

Task: Schema Design

A Ride-sharing app (like Uber) wants to analyze "Revenue per City" and "Average Trip Duration per Driver."

Problem:

1. Identify one **Fact** (quantitative) and three **Dimensions** (qualitative) for this scenario.
2. Draw a simple Star Schema linking these elements.
3. **Math Question:** If the Fact table has 1 million rows and the "City" dimension has 10 rows, explain why joining these is more efficient than having a single table with 1 million rows containing city names as strings.

Assessment (Bloom's C4 - Analysis)

Problem: The Black Friday Crash A retail company runs their Data Science reports directly on their production SQL Database (OLTP) during a Black Friday sale. The website crashes.

1. Analyze why a heavy "Mean/Standard Deviation" calculation on 1 million rows would lock a transactional database.
2. Using the "5 V's" from Week 1, argue why a **Data Warehouse** is a better architectural choice for "Velocity" and "Volume."
3. Contrast the data redundancy in a Warehouse (De-normalized) with the storage efficiency of a Database (Normalized). Which is cheaper for a CSE engineer today?

The Rise of the Data Lake

- **Problem:** Warehouses are expensive and only take structured data.
- **Solution:** A Data Lake stores data in its **raw** format (JSON, Images, Logs, CSV) until it is needed.
- **Technology Logic:** Distributed File Systems (like HDFS).
Instead of one big computer, we use 1,000 cheap ones.

Key Distinction

Warehouse = A library (Sorted, Cataloged).

Lake = A large container of water (Everything is there, but unsorted).

Schema-on-Write vs. Schema-on-Read

	Schema-on-Write	Schema-on-Read
System	Data Warehouse	Data Lake
Flexibility	Low (Hard to change)	High (Store anything)
Quality	High (Pre-cleaned)	Variable (Messy)
DS Task	Reporting / BI	Exploration / ML

- **Schema-on-Read Logic:** We define what the data "is" only when we pull it out to analyze it.

Distributed Storage Logic (Conceptual)

How do we store a 10TB file if our biggest hard drive is 1TB?

1. **Partitioning:** Split the file into 10 blocks.
2. **Replication:** Store each block on 3 different machines (for safety).
3. **Metadata:** A "NameNode" keeps track of which machine has which block.

Math of Reliability

If the probability of one machine failing is p , the probability of losing data with 3 replicas is p^3 .

Assessment (Bloom's C3 - Application)

Task: Storage Selection

Categorize where the following should be stored (Warehouse vs. Lake):

- Last year's audited financial spreadsheets.
- Raw CCTV footage from 500 cameras.
- Social media "Likes" in JSON format.
- Employee payroll records.

Problem: Pick one "Lake" item and describe the "Schema" you would apply to it when you finally decide to "Read" it for a statistical report.

Assessment (Bloom's C4 - Analysis)

The "Data Swamp" Paradox A company stops using a Warehouse and puts everything into a Data Lake. Two years later, nobody can find anything, and the data formats are inconsistent.

1. Analyze why "Metadata" is the difference between a Data Lake and a Data Swamp.
2. **Logic Question:** In a Schema-on-Read system, if the data format changes halfway through (e.g., from CSV to JSON), whose responsibility is it to fix the "Math"? The Database Administrator or the Data Scientist?
3. Compare the **Cost of Ingestion** (putting data in) vs. **Cost of Querying** (getting data out) for both architectures.